Previous-Year Mid-Semester Papers

Complete Worked Solutions

Linear Statistical Models, B.Stat. 3rd year — 2024–25 and 2025–26

Both papers are worth 40 marks. Every part is solved in full, with the alternative route given wherever the professor's phrasing admits one. Section 3 adds five worked practice questions in the same pure linear-algebra style, and Section 4 is a coverage audit against the crash courses, recording where each result now lives.

Contents

1 Mid-Semester Examination, 13 September 2024

Question 1.1. [**10** marks]

$\mathbf{P}$ and $\mathbf{Q}$ are $n \times n$ symmetric orthogonal projection matrices. Show that $\mathbf{P}+\mathbf{Q}$ is an orthogonal projection matrix if and only if $\mathbf{PQ} = \mathbf{O}_n$. In that case show also that $\operatorname{rank}(\mathbf{P}) + \operatorname{rank}(\mathbf{Q}) = \operatorname{rank}(\mathbf{P}+\mathbf{Q})$.

Solution. Recall the characterisation: M is an orthogonal projection matrix iff $M^T = M$ and $M^2 = M$. Since $\mathbf{P}, \mathbf{Q}$ are symmetric, $\mathbf{P}+\mathbf{Q}$ is symmetric automatically, so the whole question is about idempotence. Expand:

$$(\mathbf{P}+\mathbf{Q})^2 = \mathbf{P}^2 + \mathbf{PQ} + \mathbf{QP} + \mathbf{Q}^2 = \mathbf{P}+\mathbf{Q} + (\mathbf{PQ}+\mathbf{QP}),$$

so

$$(\mathbf{P}+\mathbf{Q})^2 = \mathbf{P}+\mathbf{Q} \iff \mathbf{PQ}+\mathbf{QP} = \mathbf{O}. \tag{1}$$

(⇐) Suppose $\mathbf{PQ} = \mathbf{O}$. Since $\mathbf{P}, \mathbf{Q}$ are symmetric, $\mathbf{QP} = \mathbf{Q}^T\mathbf{P}^T = (\mathbf{PQ})^T = \mathbf{O}$ as well. Hence $\mathbf{PQ}+\mathbf{QP} = \mathbf{O}$ and (1) gives idempotence.

(⇒) Suppose $\mathbf{P}+\mathbf{Q}$ is an orthogonal projector, so by (1), $\mathbf{PQ}+\mathbf{QP} = \mathbf{O}$. Multiply this on the *left* by $\mathbf{P}$ and use $\mathbf{P}^2 = \mathbf{P}$:

$$\mathbf{PQ} + \mathbf{PQP} = \mathbf{O}. \tag{i}$$

Multiply it on the *right* by $\mathbf{P}$:

$$\mathbf{PQP} + \mathbf{QP} = \mathbf{O}. \tag{ii}$$

Subtracting (ii) from (i) gives $\mathbf{PQ}-\mathbf{QP} = \mathbf{O}$, i.e. $\mathbf{PQ} = \mathbf{QP}$. Substituting back into $\mathbf{PQ}+\mathbf{QP} = \mathbf{O}$ gives $2\mathbf{PQ} = \mathbf{O}$, hence $\mathbf{PQ} = \mathbf{O}$. ■

(Rank) Now assume $\mathbf{PQ} = \mathbf{O}$, so $\mathbf{P}+\mathbf{Q}$ is idempotent. For an idempotent matrix all eigenvalues are 0 or 1, so the trace (the sum of the eigenvalues) counts the eigenvalues equal to 1, i.e. $\operatorname{tr}(M) = \operatorname{rank}(M)$. Applying this to all three idempotent matrices and using linearity of the trace:

$$\operatorname{rank}(\mathbf{P}+\mathbf{Q}) = \operatorname{tr}(\mathbf{P}+\mathbf{Q}) = \operatorname{tr}(\mathbf{P}) + \operatorname{tr}(\mathbf{Q}) = \operatorname{rank}(\mathbf{P}) + \operatorname{rank}(\mathbf{Q}). \qquad \square$$

Alternative rank argument (no trace)

Show $\mathcal{C}(\mathbf{P}+\mathbf{Q}) = \mathcal{C}(\mathbf{P}) \oplus \mathcal{C}(\mathbf{Q})$.

- *Orthogonality.* For $u = \mathbf{P}a$ and $v = \mathbf{Q}b$, $u^T v = a^T \mathbf{P}\mathbf{Q}b = 0$. So $\mathcal{C}(\mathbf{P}) \perp \mathcal{C}(\mathbf{Q})$, and in particular the sum is direct.
- "$\subseteq$": $(\mathbf{P}+\mathbf{Q})x = \mathbf{P}x + \mathbf{Q}x$.
- "$\supseteq$": $(\mathbf{P}+\mathbf{Q})\mathbf{P}a = \mathbf{P}^2 a + \mathbf{Q}\mathbf{P}a = \mathbf{P}a$, so $\mathcal{C}(\mathbf{P}) \subseteq \mathcal{C}(\mathbf{P}+\mathbf{Q})$; similarly for $\mathbf{Q}$.

Dimensions of a direct sum add, giving the result.

Question 1.2. [**12** marks]

Data $\mathbf{Y} = \{Y_{ij} : 1 \le j \le m,\ i = 1,2,3,4\}$, $m \ge 2$, index i the treatment and j the replicate, with

$$Y_{1j} = \beta_1 + \varepsilon_{1j}, \quad Y_{2j} = \beta_1 + \beta_2 + \varepsilon_{2j}, \quad Y_{3j} = \beta_2 + \beta_3 + \varepsilon_{3j}, \quad Y_{4j} = \beta_3 + \beta_4 + \varepsilon_{4j},$$

$\varepsilon_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. **(a)** Write the model in matrix form. **(b)** Are (i) β_2; (ii) $\beta_1 - \beta_2$; (iii) β_4 estimable? For each estimable one find the BLUE. **(c)** Propose a test of $H_0 : \beta_1 = \beta_2$ against $H_A : \beta_1 \ne \beta_2$.

(a) Stack $\mathbf{Y}$ treatment-major, $\mathbf{Y} = (Y_{11}, \dots, Y_{1m}, Y_{21}, \dots, Y_{2m}, Y_{31}, \dots, Y_{4m})^T$, and put $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3, \beta_4)^T$. Then $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}_{4m}(\mathbf{0}, \sigma^2 I_{4m})$ and

$$\underset{4m\times 4}{\mathbf{X}} = \begin{pmatrix} \mathbf{1}_m & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1}_m & \mathbf{1}_m & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_m & \mathbf{1}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1}_m & \mathbf{1}_m \end{pmatrix} = A \otimes \mathbf{1}_m, \qquad A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

(b) First compute the rank — do not assume it is deficient. A is lower triangular with all diagonal entries equal to 1, so $\det A = 1$ and A is invertible. Hence $\operatorname{rank}(\mathbf{X}) = \operatorname{rank}(A) = 4 = p$: the model has **full column rank**, so $\mathbf{X}^T\mathbf{X} = m\, A^T A$ is non-singular, the model is identifiable, and

Every linear function of β is estimable

All three of β_2, $\beta_1 - \beta_2$ and β_4 are estimable. This is the trap in the question: the model *looks* like an over-parametrised ANOVA, but it is not.

Write $\mu_i := \mathbb{E}(Y_{ij})$, so $\mu_1 = \beta_1$, $\mu_2 = \beta_1 + \beta_2$, $\mu_3 = \beta_2 + \beta_3$, $\mu_4 = \beta_3 + \beta_4$, i.e. $\boldsymbol{\mu} = A\boldsymbol{\beta}$. Since $\mathcal{C}(\mathbf{X})$ is exactly the span of the four treatment indicators (because A is invertible), the LSE of $\boldsymbol{\mu}$ is the vector of group means and, by forward substitution in $A\boldsymbol{\beta} = \boldsymbol{\mu}$,

$$\widehat{\boldsymbol{\beta}} = A^{-1}\begin{pmatrix} \overline{Y}_{1\cdot} \\ \overline{Y}_{2\cdot} \\ \overline{Y}_{3\cdot} \\ \overline{Y}_{4\cdot} \end{pmatrix} : \qquad \begin{aligned} \widehat{\beta}_1 &= \overline{Y}_{1\cdot}, \\ \widehat{\beta}_2 &= \overline{Y}_{2\cdot} - \overline{Y}_{1\cdot}, \end{aligned} \qquad \begin{aligned} \widehat{\beta}_3 &= \overline{Y}_{3\cdot} - \overline{Y}_{2\cdot} + \overline{Y}_{1\cdot}, \\ \widehat{\beta}_4 &= \overline{Y}_{4\cdot} - \overline{Y}_{3\cdot} + \overline{Y}_{2\cdot} - \overline{Y}_{1\cdot}. \end{aligned}$$

(Check directly: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} = (mA^TA)^{-1}(A^T \otimes \mathbf{1}_m^T)\mathbf{Y} = (A^TA)^{-1}A^T\bar{\mathbf{Y}} = A^{-1}\bar{\mathbf{Y}}$, where $\bar{\mathbf{Y}}$ is the vector of group means.)

The $\overline{Y}_{i\cdot}$ are independent with variance σ^2/m, so for a coefficient vector c the variance is $\frac{\sigma^2}{m}\sum_i c_i^2$. By Gauss–Markov each of these is the unique BLUE:

	Parameter	BLUE	Variance
(i)	β_2	$\overline{Y}_{2\cdot} - \overline{Y}_{1\cdot}$	$2\sigma^2/m$
(ii)	$\beta_1 - \beta_2 = 2\mu_1 - \mu_2$	$2\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot}$	$5\sigma^2/m$
(iii)	β_4	$\overline{Y}_{4\cdot} - \overline{Y}_{3\cdot} + \overline{Y}_{2\cdot} - \overline{Y}_{1\cdot}$	$4\sigma^2/m$

(c) Write the hypothesis as $L^T\boldsymbol{\beta} = \theta_0$ with $L^T = (1, -1, 0, 0)$, $\theta_0 = 0$, so $r = \mathrm{rank}(L) = 1$. Then

$$\widehat{\theta} = \widehat{\beta}_1 - \widehat{\beta}_2 = 2\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot}, \qquad \mathrm{Var}(\widehat{\theta}) = \frac{5\sigma^2}{m}.$$

Because $\mathcal{C}(\mathbf{X})$ is the span of the group indicators, the fitted values are $\widehat{Y}_{ij} = \overline{Y}_{i\cdot}$ and

$$\mathrm{SSE} = \sum_{i=1}^{4}\sum_{j=1}^{m}(Y_{ij} - \overline{Y}_{i\cdot})^2, \qquad \widehat{\sigma}^2 = \mathrm{MSE} = \frac{\mathrm{SSE}}{4m-4}, \qquad \mathrm{SSE} \sim \sigma^2\chi^2_{4m-4} \text{ independently of } \widehat{\boldsymbol{\beta}}.$$

Hence the test statistic

$$\boxed{T = \frac{2\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot}}{\widehat{\sigma}\sqrt{5/m}} \overset{H_0}{\sim} t_{4m-4}}, \qquad \text{reject } H_0 \text{ if } |T| > t^{1-\alpha/2}_{4m-4}.$$

Equivalently (and this is the "alternative form of the likelihood ratio test" the note permits) use

$$F = T^2 = \frac{W}{r} = \frac{(\mathrm{SSE}_{red} - \mathrm{SSE}_{full})/1}{\mathrm{MSE}} \overset{H_0}{\sim} F_{1,4m-4},$$

where SSE_{red} is the residual sum of squares after fitting under the constraint $\beta_1 = \beta_2$.

Question 1.3. [**6** marks]

> A is $m \times m$ of the form $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ with A_{11} of size $r \times r$, $r < m$, and $\mathrm{rank}(A) = \mathrm{rank}(A_{11}) = r$. Show that $G = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$ is a generalised inverse of A.

Solution. A_{11} is $r \times r$ with $\mathrm{rank}(A_{11}) = r$, so A_{11}^{-1} exists and G is well defined. We must verify $AGA = A$. First,

$$AG = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}\begin{pmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{pmatrix},$$

and therefore

$$AGA = \begin{pmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{pmatrix}\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{pmatrix}. \tag{2}$$

Comparing with A, it remains *only* to show

$$A_{22} = A_{21}A_{11}^{-1}A_{12}, \tag{3}$$

i.e. that the Schur complement $A_{22} - A_{21}A_{11}^{-1}A_{12}$ vanishes.

This is exactly what the rank hypothesis gives. Write A in terms of its two row blocks, $A = \begin{bmatrix} [A_{11} \ A_{12}] \\ [A_{21} \ A_{22}] \end{bmatrix}$. The top block $[A_{11} \ A_{12}]$ has r rows and already has rank r (since A_{11} does), and $\mathrm{rank}(A) = r$ by hypothesis. So the row space of A is spanned by the first r rows, and every row of the bottom block is a linear combination of them: there is an $(m-r) \times r$ matrix M with

$$[A_{21} \quad A_{22}] = M[A_{11} \quad A_{12}] \quad \Longrightarrow \quad A_{21} = MA_{11} \ \text{ and } \ A_{22} = MA_{12}.$$

Since A_{11} is invertible, the first equation gives $M = A_{21}A_{11}^{-1}$, and substituting into the second gives (3). With (3), (2) reads $AGA = A$. ■

Common confusion: where the hypothesis $\operatorname{rank}(A) = r$ is used

Only in (3). If you drop it, G still satisfies AG idempotent, but AGA differs from A in the bottom-right block — so G would *not* be a g-inverse. Say explicitly where you use the rank condition; that is where the marks are.

Question 1.4. [**12** marks]

$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{X}$ is $n \times p$ with $\operatorname{rank}(\mathbf{X}) = p$, $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I_n)$. $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_1, \dots, \widehat{\beta}_p)^T$ is the LSE and $\widehat{\sigma}^2 = \text{MSE}$. Let $\mathbf{X}_{\cdot j}$ be the j-th column and $\mathbf{X}_{-j}$ the $n \times (p-1)$ submatrix with the j-th column removed. **(a)** Show $\operatorname{Var}(\widehat{\beta}_j) = \sigma^2 / \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j}$. **(b)** Show the t-statistic for $H_0 : \beta_j = 0$ can be written $T_j = \frac{1}{\widehat{\sigma}/\sigma} \big(\frac{\beta_j}{\sigma} \sqrt{\mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j}} + W_j \big)$ with $W_j \sim \mathcal{N}(0,1)$ independent of $\widehat{\sigma}^2$. **(c)** If $\mathbb{E}(\mathbf{Y}) = \beta_{j_0} \mathbf{X}_{\cdot j_0}$ and $\mathbf{X}_{\cdot j_0}$ is nearly collinear with the other columns, argue that the t-test in the full model is much less significant than in the true model.

Throughout write $d_j := \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} = \|(I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j}\|^2$.

(a) Apply Frisch–Waugh with $\mathbf{Z} = \mathbf{X}_{-j}$ (the other $p-1$ columns) and $\mathbf{W} = \mathbf{X}_{\cdot j}$ (one column, coefficient β_j):

$$\widehat{\beta}_j = \big(\mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} \big)^{-1} \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{Y} = \frac{1}{d_j} \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{Y}.$$

So $\widehat{\beta}_j = a^T \mathbf{Y}$ with $a = (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} / d_j$, and since $\operatorname{Var}(\mathbf{Y}) = \sigma^2 I$ and $I - \mathbf{P}_{-j}$ is symmetric idempotent,

$$\operatorname{Var}(\widehat{\beta}_j) = \sigma^2 a^T a = \frac{\sigma^2}{d_j^2} \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j})(I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} = \frac{\sigma^2}{d_j^2} \cdot d_j = \frac{\sigma^2}{d_j}. \qquad \square$$

Second route: Schur complement

$\operatorname{Var}(\widehat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$, so $\operatorname{Var}(\widehat{\beta}_j) = \sigma^2 [(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}$. Permute the j-th column to the end and use the partitioned inverse: the bottom-right entry of $\begin{bmatrix} \mathbf{X}_{-j}^T \mathbf{X}_{-j} & \mathbf{X}_{-j}^T \mathbf{X}_{\cdot j} \\ \mathbf{X}_{\cdot j}^T \mathbf{X}_{-j} & \mathbf{X}_{\cdot j}^T \mathbf{X}_{\cdot j} \end{bmatrix}^{-1}$ is the reciprocal of the Schur complement $\mathbf{X}_{\cdot j}^T \mathbf{X}_{\cdot j} - \mathbf{X}_{\cdot j}^T \mathbf{X}_{-j} (\mathbf{X}_{-j}^T \mathbf{X}_{-j})^{-1} \mathbf{X}_{-j}^T \mathbf{X}_{\cdot j} = \mathbf{X}_{\cdot j}^T (I - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} = d_j$.

(b) From (a), $\widehat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2 / d_j)$. Standardising,

$$W_j := \frac{(\widehat{\beta}_j - \beta_j)\sqrt{d_j}}{\sigma} \sim \mathcal{N}(0,1) \qquad \Longrightarrow \qquad \frac{\widehat{\beta}_j \sqrt{d_j}}{\sigma} = \frac{\beta_j \sqrt{d_j}}{\sigma} + W_j.$$

The t-statistic is $T_j = \widehat{\beta}_j / s(\widehat{\beta}_j)$ with $s(\widehat{\beta}_j) = \widehat{\sigma} / \sqrt{d_j}$, so, inserting σ/σ,

$$T_j = \frac{\widehat{\beta}_j \sqrt{d_j}}{\widehat{\sigma}} = \frac{\sigma}{\widehat{\sigma}} \cdot \frac{\widehat{\beta}_j \sqrt{d_j}}{\sigma} = \frac{1}{\widehat{\sigma}/\sigma} \left(\frac{\beta_j}{\sigma} \sqrt{d_j} + W_j \right).$$

Finally W_j is a function of $\widehat{\boldsymbol{\beta}}$ alone, and $\widehat{\boldsymbol{\beta}} \perp\!\!\!\perp \text{SSE}$ (shown by $\mathbb{E}[(I - \mathbf{P}_{\mathbf{X}}) \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}] = \sigma^2 (I - \mathbf{P}_{\mathbf{X}}) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} = 0$ plus joint normality), so $W_j \perp\!\!\!\perp \widehat{\sigma}^2$. ■

Note that under $H_0 : \beta_j = 0$ this is $T_j = W_j / \sqrt{\widehat{\sigma}^2 / \sigma^2}$ with $(n-p)\widehat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p}$, i.e. exactly t_{n-p}.

(c) Part (b) shows the whole signal in T_{j_0} sits in the **non-centrality**

$$\frac{\beta_{j_0}}{\sigma} \sqrt{d_{j_0}}, \qquad d_{j_0} = \|(I - \mathbf{P}_{-j_0}) \mathbf{X}_{\cdot j_0}\|^2.$$

- **Full model.** "$\mathbf{X}_{\cdot j_0}$ nearly collinear with the other columns" means $\mathbf{X}_{\cdot j_0}$ is close to $\mathcal{C}(\mathbf{X}_{-j_0})$, so $(I - \mathbf{P}_{-j_0})\mathbf{X}_{\cdot j_0} \approx \mathbf{0}$ and $d_{j_0} \approx 0$. Then the non-centrality is ≈ 0 and $T_{j_0} \approx W_{j_0}/(\widehat{\sigma}/\sigma)$, which is (approximately) a *central* t_{n-p} — the same distribution as under H_0. The test therefore has power close to its level: it will typically fail to reject even though $\beta_{j_0} \neq 0$.
- **True model** $\mathbf{Y} \sim \mathcal{N}_n(\beta_{j_0}\mathbf{X}_{\cdot j_0}, \sigma^2 I)$. Here there are no other columns to project out, so $d = \|\mathbf{X}_{\cdot j_0}\|^2$, which is large. The non-centrality $\beta_{j_0}\|\mathbf{X}_{\cdot j_0}\|/\sigma$ is large and the test is highly significant. (The error variance also gets $n-1$ degrees of freedom instead of $n-p$.)

Equivalently, $\mathrm{Var}(\widehat{\beta}_{j_0}) = \sigma^2/d_{j_0}$ is inflated by collinearity. Note carefully that $\widehat{\beta}_{j_0}$ remains **unbiased** in the full model — collinearity destroys precision and hence power, it does not introduce bias.

2 Mid-Semester Examination, 9 September 2025

Question 2.1. [**5** marks]

$\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$, $\boldsymbol{\beta} \in \mathbb{R}^p$, $\mathrm{rank}(\mathbf{X}) = p$, $\mathbf{X} = [\mathbf{Z} : \mathbf{W}]$ with $\mathbf{Z}$ an $n \times q$ matrix, $1 \le q < p$. Define $S_1 = \min_{\boldsymbol{\beta}\in\mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$ and $S_0 = \min_{\gamma\in\mathbb{R}^q} \|\mathbf{Y} - \mathbf{Z}\gamma\|_2^2$. Show $S_1/S_0 \sim \mathrm{Beta}(k_1, k_2)$ with $k_1 = (n-p)/2$ and $k_2 = (p-q)/2$, provided $\mathbb{E}(\mathbf{Y}) \in \mathcal{C}(\mathbf{Z})$.

Solution. Both minima are squared lengths of residual vectors, i.e. quadratic forms in orthogonal projections:

$$S_1 = \|(I - \mathbf{P_X})\mathbf{Y}\|^2 = \mathbf{Y}^T(I - \mathbf{P_X})\mathbf{Y}, \qquad S_0 = \|(I - \mathbf{P_Z})\mathbf{Y}\|^2 = \mathbf{Y}^T(I - \mathbf{P_Z})\mathbf{Y}.$$

Write $S_0 = S_1 + (S_0 - S_1)$ where, since $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$,

$$S_0 - S_1 = \mathbf{Y}^T\big(\mathbf{P_X} - \mathbf{P_Z}\big)\mathbf{Y} = \mathbf{Y}^T\mathbf{P_{W|Z}}\mathbf{Y}.$$

Put $A_1 = I - \mathbf{P_X}$ and $A_2 = \mathbf{P_X} - \mathbf{P_Z}$. Both are symmetric and idempotent (A_2 by the decomposition lemma $\mathbf{P_X} = \mathbf{P_Z} + \mathbf{P_{W|Z}}$), with

$$\mathrm{rank}(A_1) = n - \mathrm{rank}(\mathbf{X}) = n - p, \qquad \mathrm{rank}(A_2) = \mathrm{rank}(\mathbf{X}) - \mathrm{rank}(\mathbf{Z}) = p - q.$$

Independence. Using $\mathbf{P_X P_Z} = \mathbf{P_Z P_X} = \mathbf{P_Z}$,

$$A_1A_2 = (I - \mathbf{P_X})(\mathbf{P_X} - \mathbf{P_Z}) = \mathbf{P_X} - \mathbf{P_Z} - \mathbf{P_X} + \mathbf{P_Z} = O.$$

Two quadratic forms in a normal vector with $A_1\Sigma A_2 = \sigma^2 A_1A_2 = O$ are independent.

Both are central. By hypothesis $\boldsymbol{\mu} := \mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{Z})$, so $\mathbf{P_Z}\boldsymbol{\mu} = \boldsymbol{\mu}$ and also $\mathbf{P_X}\boldsymbol{\mu} = \boldsymbol{\mu}$. Hence

$$A_1\boldsymbol{\mu} = \boldsymbol{\mu} - \boldsymbol{\mu} = \mathbf{0}, \qquad A_2\boldsymbol{\mu} = \boldsymbol{\mu} - \boldsymbol{\mu} = \mathbf{0},$$

so both non-centrality parameters vanish and, by the quadratic-form theorem,

$$\frac{S_1}{\sigma^2} \sim \chi^2_{n-p}, \qquad \frac{S_0 - S_1}{\sigma^2} \sim \chi^2_{p-q}, \qquad \text{independently.}$$

Beta step. Write $A = S_1/\sigma^2$ and $B = (S_0 - S_1)/\sigma^2$. Then

$$\frac{S_1}{S_0} = \frac{S_1}{S_1 + (S_0 - S_1)} = \frac{A}{A+B}.$$

If $A \sim \chi^2_a = \mathrm{Gamma}(a/2, 2)$ and $B \sim \chi^2_b = \mathrm{Gamma}(b/2, 2)$ are independent, the change of variables $(U, V) = \big(\frac{A}{A+B}, A + B\big)$ has Jacobian v and factorises the joint density into $u^{a/2-1}(1-u)^{b/2-1}$ times a $\mathrm{Gamma}(\frac{a+b}{2}, 2)$ density; hence $U \sim \mathrm{Beta}(a/2, b/2)$ independently of V. With $a = n - p$ and $b = p - q$,

$$\boxed{\frac{S_1}{S_0} \sim \mathrm{Beta}\Big(\frac{n-p}{2}, \frac{p-q}{2}\Big).} \qquad \square$$

Common confusion: which way up?

$S_1 \le S_0$ always (the bigger model fits at least as well), so the ratio S_1/S_0 lies in $[0,1]$ — as a Beta variate must. If you accidentally write S_0/S_1 you get a quantity ≥ 1, which cannot be Beta. This is the fastest sanity check on the orientation, and it is also why the first parameter is $(n-p)/2$ (the *numerator's* degrees of freedom).

Question 2.2. [**15** marks]

$Y_{ijk} = \alpha_i + \beta_j + \varepsilon_{ijk}$, $1 \le i,j \le 2$, $k = 1,\dots,m$, $\varepsilon_{ijk} \overset{\text{iid}}{\sim} \mathcal{N}(0,\sigma^2)$, $m \ge 2$. **(a)** [3] Show the design matrix $\mathbf{X}$ has rank 3. **(b)** [4] Find the BLUE of $\theta = \mathbf{X}^T\boldsymbol{\mu}$, where $\boldsymbol{\mu} = \mathbb{E}(\mathbf{Y})$ has entries $\mu_{ij} = \mathbb{E}(Y_{ijk})$. **(c)** [8] For each of (i) α_1; (ii) $\alpha_1-\alpha_2$; (iii) $\alpha_1+\alpha_2+\beta_1+\beta_2$; (iv) $\alpha_1-\alpha_2+\beta_1-\beta_2$, check estimability and, if estimable, find the BLUE. Justify.

Order the parameters as $\boldsymbol{\beta} = (\alpha_1,\alpha_2,\beta_1,\beta_2)^T$ and the observations by cell $(1,1),(1,2),(2,1),(2,2)$, each cell contributing m identical rows. The four distinct rows of $\mathbf{X}$ are

$$(1,0,1,0), \quad (1,0,0,1), \quad (0,1,1,0), \quad (0,1,0,1),$$

so $\mathbf{X} = A \otimes \mathbf{1}_m$ with A the 4×4 matrix of those rows.

(a) Upper bound. Writing $c_1,\dots,c_4$ for the columns of $\mathbf{X}$, both c_1+c_2 and c_3+c_4 equal $\mathbf{1}_{4m}$ (every observation has exactly one α and exactly one β). Hence

$$c_1+c_2-c_3-c_4 = \mathbf{0} \quad\Longrightarrow\quad \operatorname{rank}(\mathbf{X}) \le 3.$$

Lower bound. The rows $(1,0,1,0)$, $(1,0,0,1)$, $(0,1,1,0)$ are linearly independent: if $a(1,0,1,0)+b(1,0,0,1)+c(0,1,1,0) = \mathbf{0}$ then the second coordinate gives $c=0$, the fourth gives $b=0$, and the first then gives $a=0$. Hence $\operatorname{rank}(\mathbf{X}) \ge 3$. Therefore $\operatorname{rank}(\mathbf{X}) = 3$. ■

Consequently $\dim \mathcal{N}(\mathbf{X}) = 4-3 = 1$ and, from the dependency above,

$$\mathcal{N}(\mathbf{X}) = \operatorname{span}\{z\}, \qquad z = (1,1,-1,-1)^T. \tag{4}$$

(The model is not identifiable: replacing $\alpha_i \mapsto \alpha_i + c$, $\beta_j \mapsto \beta_j - c$ leaves every μ_{ij} unchanged.)

(b) Since $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, we have $\theta = \mathbf{X}^T\boldsymbol{\mu} = \mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$. Each coordinate of θ is $a^T\boldsymbol{\beta}$ with $a = \mathbf{X}^T\mathbf{X}e_k \in \mathcal{C}(\mathbf{X}^T\mathbf{X})$, so θ **is estimable without any condition**. Its BLUE is $a^T\widehat{\boldsymbol{\beta}}$ evaluated at any solution of the normal equations, i.e. $\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{Y}$, giving

$$\boxed{\widehat{\theta} = \mathbf{X}^T\mathbf{Y}} \qquad (\text{equivalently } \mathbf{X}^T\widehat{\boldsymbol{\mu}} = \mathbf{X}^T\mathbf{P}_\mathbf{X}\mathbf{Y} = (\mathbf{P}_\mathbf{X}\mathbf{X})^T\mathbf{Y} = \mathbf{X}^T\mathbf{Y}).$$

Explicitly, $\mathbf{X}^T\mathbf{Y}$ collects the row and column totals:

$$\widehat{\theta} = \big(T_{1\cdot\cdot},\ T_{2\cdot\cdot},\ T_{\cdot 1\cdot},\ T_{\cdot 2\cdot}\big)^T, \qquad T_{i\cdot\cdot} = \sum_{j,k} Y_{ijk}, \quad T_{\cdot j\cdot} = \sum_{i,k} Y_{ijk},$$

with $\operatorname{Var}(\widehat{\theta}) = \sigma^2\mathbf{X}^T\mathbf{X} = \sigma^2 m \begin{bmatrix} 2&0&1&1\\ 0&2&1&1\\ 1&1&2&0\\ 1&1&0&2 \end{bmatrix}$.

(c) By (4) and the criterion $a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{N}(\mathbf{X})^\perp$,

$$a^T\boldsymbol{\beta} \text{ is estimable} \iff a^Tz = 0 \iff a_1+a_2 = a_3+a_4. \tag{5}$$

For the BLUEs, note that in this balanced additive design the LS fitted cell means are

$$\widehat{\mu}_{ij} = \overline{Y}_{i\cdot\cdot} + \overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdot\cdot\cdot},$$

and any estimable function is a linear combination of the μ_{ij}, whose BLUE is the same combination of the $\widehat{\mu}_{ij}$. Write $M_{ij} := \overline{Y}_{ij\cdot}$ for the four cell means; they are independent with variance σ^2/m.

(i) α_1: $a = (1, 0, 0, 0)$, $a^T z = 1 \neq 0$. **Not estimable.**
Justification in words: for any c, the parameter vectors $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + cz$ give identical distributions for $\mathbf{Y}$, but different values of α_1; so no statistic can have expectation α_1 for all $\boldsymbol{\beta}$.
(ii) $\alpha_1 - \alpha_2$: $a = (1, -1, 0, 0)$, $a^T z = 1 - 1 = 0$. **Estimable.** It equals $\mu_{1j} - \mu_{2j}$ for either j, so

$$\widehat{\alpha_1 - \alpha_2} = \widehat{\mu}_{11} - \widehat{\mu}_{21} = \big(\overline{Y}_{1\cdot\cdot} + \overline{Y}_{\cdot 1\cdot} - \overline{Y}_{\cdot\cdot\cdot}\big) - \big(\overline{Y}_{2\cdot\cdot} + \overline{Y}_{\cdot 1\cdot} - \overline{Y}_{\cdot\cdot\cdot}\big) = \boxed{\overline{Y}_{1\cdot\cdot} - \overline{Y}_{2\cdot\cdot}}.$$

$\overline{Y}_{1\cdot\cdot}$ and $\overline{Y}_{2\cdot\cdot}$ are means of $2m$ observations each, over disjoint sets, hence independent with variance $\sigma^2/(2m)$; so $\mathrm{Var} = \sigma^2/m$.
(iii) $\alpha_1 + \alpha_2 + \beta_1 + \beta_2$: $a = (1, 1, 1, 1)$, $a^T z = 1 + 1 - 1 - 1 = 0$. **Estimable.** It equals $(\alpha_1 + \beta_1) + (\alpha_2 + \beta_2) = \mu_{11} + \mu_{22}$ (and also $\mu_{12} + \mu_{21}$). Then

$$\widehat{\mu}_{11} + \widehat{\mu}_{22} = \big(\overline{Y}_{1\cdot\cdot} + \overline{Y}_{2\cdot\cdot}\big) + \big(\overline{Y}_{\cdot 1\cdot} + \overline{Y}_{\cdot 2\cdot}\big) - 2\overline{Y}_{\cdot\cdot\cdot} = 2\overline{Y}_{\cdot\cdot\cdot} + 2\overline{Y}_{\cdot\cdot\cdot} - 2\overline{Y}_{\cdot\cdot\cdot} = \boxed{2\overline{Y}_{\cdot\cdot\cdot}},$$

using $\overline{Y}_{1\cdot\cdot} + \overline{Y}_{2\cdot\cdot} = \overline{Y}_{\cdot 1\cdot} + \overline{Y}_{\cdot 2\cdot} = 2\overline{Y}_{\cdot\cdot\cdot}$ in the balanced design. $\mathrm{Var} = 4 \cdot \sigma^2/(4m) = \sigma^2/m$.
(iv) $\alpha_1 - \alpha_2 + \beta_1 - \beta_2$: $a = (1, -1, 1, -1)$, $a^T z = 1 - 1 - 1 + 1 = 0$. **Estimable.** It equals $(\alpha_1 + \beta_1) - (\alpha_2 + \beta_2) = \mu_{11} - \mu_{22}$, so

$$\widehat{\mu}_{11} - \widehat{\mu}_{22} = \big(\overline{Y}_{1\cdot\cdot} - \overline{Y}_{2\cdot\cdot}\big) + \big(\overline{Y}_{\cdot 1\cdot} - \overline{Y}_{\cdot 2\cdot}\big) = \boxed{(\text{row contrast}) + (\text{column contrast})}.$$

In terms of the cell means, the row contrast is $R = \frac{1}{2}(M_{11} + M_{12} - M_{21} - M_{22})$ and the column contrast is $C = \frac{1}{2}(M_{11} - M_{12} + M_{21} - M_{22})$; each has variance $\frac{1}{4} \cdot 4 \cdot \frac{\sigma^2}{m} = \frac{\sigma^2}{m}$, and

$$\mathrm{Cov}(R, C) = \tfrac{1}{4} \cdot \tfrac{\sigma^2}{m}\big[(1)(1) + (1)(-1) + (-1)(1) + (-1)(-1)\big] = 0,$$

so row and column contrasts are uncorrelated (orthogonality of the balanced design) and $\mathrm{Var} = 2\sigma^2/m$.

Question 2.3. [**12** marks]

$Y_{ij} = \mu_i + \beta_1 z_{ij} + \beta_2 w_{ij} + \varepsilon_{ij}$, $j = 1, \ldots, n_i$, $i = 1, \ldots, r$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, $n = \sum_i n_i$. With $\bar{z}_{i\cdot} = \frac{1}{n_i}\sum_j z_{ij}$ etc.,

$$S_{zz} = \sum_i \sum_j (z_{ij} - \bar{z}_{i\cdot})^2, \quad S_{zw} = \sum_i \sum_j (z_{ij} - \bar{z}_{i\cdot})(w_{ij} - \bar{w}_{i\cdot}), \quad S_{ww} = \sum_i \sum_j (w_{ij} - \bar{w}_{i\cdot})^2,$$

$$S_{zy} = \sum_i \sum_j (z_{ij} - \bar{z}_{i\cdot})(Y_{ij} - \overline{Y}_{i\cdot}), \qquad S_{wy} = \sum_i \sum_j (w_{ij} - \bar{w}_{i\cdot})(Y_{ij} - \overline{Y}_{i\cdot}),$$

and $\widehat{\sigma}^2 = \mathrm{MSE}$. **(a)** [4] Show $\widehat{\boldsymbol{\beta}} = \left[\begin{smallmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{smallmatrix}\right]^{-1} \left[\begin{smallmatrix} S_{zy} \\ S_{wy} \end{smallmatrix}\right]$ for $\boldsymbol{\beta} = (\beta_1, \beta_2)^T$. **(b)** [4] Show the LRT of $H_0 : \beta_2 = 0$ rejects for large $U = \big(S_{ww} - S_{zw}^2/S_{zz}\big)\widehat{\beta}_2^2/\widehat{\sigma}^2$. **(c)** [4] Show the LRT of $H_0 : \beta_1 = \beta_2 = 0$ rejects for large $V = \frac{1}{\widehat{\sigma}^2}\widehat{\boldsymbol{\beta}}^T \left[\begin{smallmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{smallmatrix}\right] \widehat{\boldsymbol{\beta}}$.

Write the model as $\mathbf{Y} = \mathbf{Z}\gamma + \mathbf{W}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{Z}$ is the $n \times r$ matrix of treatment indicators, $\gamma = (\mu_1, \ldots, \mu_r)^T$, and $\mathbf{W} = [\, z : w \,]$ is $n \times 2$. The full model has $p = r + 2$ parameters, so MSE has $n - r - 2$ degrees of freedom.

The one observation that makes all three parts easy

$\mathbf{P_Z}$ replaces each observation by its *treatment mean*, so $I - \mathbf{P_Z}$ **centres within treatment**:

$$\big[(I - \mathbf{P_Z})v\big]_{ij} = v_{ij} - \bar{v}_{i\cdot}.$$

Every S in the question is therefore an inner product of two such centred vectors, and since

$I - \mathbf{P_Z}$ is symmetric idempotent,

$$u^T(I - \mathbf{P_Z})v = \big((I - \mathbf{P_Z})u\big)^T\big((I - \mathbf{P_Z})v\big) = \sum_i\sum_j(u_{ij} - \bar{u}_{i\cdot})(v_{ij} - \bar{v}_{i\cdot}).$$

(a) By Frisch–Waugh applied with $\mathbf{Z}$ partialled out,

$$\widehat{\boldsymbol{\beta}} = \big(\mathbf{W}^T(I - \mathbf{P_Z})\mathbf{W}\big)^{-1}\mathbf{W}^T(I - \mathbf{P_Z})\mathbf{Y}.$$

By the boxed identity,

$$\mathbf{W}^T(I - \mathbf{P_Z})\mathbf{W} = \begin{pmatrix} z^T(I - \mathbf{P_Z})z & z^T(I - \mathbf{P_Z})w \\ w^T(I - \mathbf{P_Z})z & w^T(I - \mathbf{P_Z})w \end{pmatrix} = \begin{pmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{pmatrix} =: M,$$

$$\mathbf{W}^T(I - \mathbf{P_Z})\mathbf{Y} = \begin{pmatrix} z^T(I - \mathbf{P_Z})\mathbf{Y} \\ w^T(I - \mathbf{P_Z})\mathbf{Y} \end{pmatrix} = \begin{pmatrix} S_{zy} \\ S_{wy} \end{pmatrix},$$

which gives $\widehat{\boldsymbol{\beta}} = M^{-1}(S_{zy}, S_{wy})^T$. ■ Also $\operatorname{Var}(\widehat{\boldsymbol{\beta}}) = \sigma^2 M^{-1}$.

(b) This is $H_0 : L^T\boldsymbol{\beta}_{\text{full}} = 0$ with a single constraint ($r = 1$) picking out β_2. The likelihood ratio test is equivalent to the F/Wald test (reject for small $\Lambda = (\text{SSE}_{full}/\text{SSE}_{red})^{n/2} \iff$ reject for large $(\text{SSE}_{red} - \text{SSE}_{full})/\text{SSE}_{full} \iff$ reject for large F), and with one constraint

$$F = W = \frac{\widehat{\beta}_2^2}{\widehat{\sigma}^2\,[M^{-1}]_{22}}.$$

Now invert the 2×2 matrix:

$$M^{-1} = \frac{1}{S_{zz}S_{ww} - S_{zw}^2}\begin{pmatrix} S_{ww} & -S_{zw} \\ -S_{zw} & S_{zz} \end{pmatrix} \implies [M^{-1}]_{22} = \frac{S_{zz}}{S_{zz}S_{ww} - S_{zw}^2} = \frac{1}{S_{ww} - \dfrac{S_{zw}^2}{S_{zz}}}.$$

Therefore

$$\boxed{F = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}}\right)\frac{\widehat{\beta}_2^2}{\widehat{\sigma}^2} = U \overset{H_0}{\sim} F_{1,\,n-r-2}}$$

and the LRT rejects H_0 for large U. ■

Reading the coefficient

$S_{ww} - S_{zw}^2/S_{zz} = w^T(I - \mathbf{P}_{[\mathbf{Z}:z]})w$ is the residual sum of squares of w after removing *both* the treatment means and z. So U is exactly $\widehat{\beta}_2^2 \times (\text{information about } \beta_2)/\widehat{\sigma}^2$: the more of w that survives adjustment for z, the more powerful the test — the same collinearity phenomenon as 2024 Q4(c).

(c) Now $r = 2$ constraints, $L^T\boldsymbol{\beta}_{\text{full}} = \boldsymbol{\beta} = (\beta_1, \beta_2)^T = \mathbf{0}$. The Wald statistic is

$$W = (\widehat{\boldsymbol{\beta}} - \mathbf{0})^T\big(\widehat{\operatorname{Var}}(\widehat{\boldsymbol{\beta}})\big)^{-1}(\widehat{\boldsymbol{\beta}} - \mathbf{0}) = \widehat{\boldsymbol{\beta}}^T\big(\widehat{\sigma}^2 M^{-1}\big)^{-1}\widehat{\boldsymbol{\beta}} = \frac{1}{\widehat{\sigma}^2}\,\widehat{\boldsymbol{\beta}}^T M\,\widehat{\boldsymbol{\beta}} = V,$$

and the LRT rejects for large V, with

$$\boxed{\frac{V}{2} \overset{H_0}{\sim} F_{2,\,n-r-2}.}$$

Equivalently $V = (\text{SSE}_{red} - \text{SSE}_{full})/\widehat{\sigma}^2$, where SSE_{red} is the residual sum of squares of the pure one-way ANOVA model $Y_{ij} = \mu_i + \varepsilon_{ij}$. ■

Question 2.4. [**8** marks]

$\mathbf{A}$ and $\mathbf{B}$ are symmetric $n \times n$ matrices, $\mathbf{B}$ is idempotent, and $\mathbf{A}$ and $I_n - \mathbf{A} - \mathbf{B}$ are non-negative definite. Show $\mathbf{AB} = \mathbf{BA} = \mathbf{O}_n$. *[Hint: show $y^T\mathbf{A}y = 0$ for all $y \in \mathcal{C}(\mathbf{B})$.]*

Solution. **Step 1: B acts as the identity on its own column space.** $\mathbf{B}$ is symmetric and idempotent, hence the orthogonal projector onto $\mathcal{C}(\mathbf{B})$. So for every $y \in \mathcal{C}(\mathbf{B})$, writing $y = \mathbf{B}u$,

$$\mathbf{B}y = \mathbf{B}^2u = \mathbf{B}u = y \qquad \Longrightarrow \qquad y^T\mathbf{B}y = y^Ty.$$

Step 2: squeeze $y^T\mathbf{A}y$ between 0 and 0. Let $y \in \mathcal{C}(\mathbf{B})$. Non-negative definiteness of $I - \mathbf{A} - \mathbf{B}$ gives

$$0 \;\le\; y^T(I - \mathbf{A} - \mathbf{B})y \;=\; y^Ty - y^T\mathbf{A}y - \underbrace{y^T\mathbf{B}y}_{=y^Ty} \;=\; -y^T\mathbf{A}y,$$

so $y^T\mathbf{A}y \le 0$. But $\mathbf{A} \succeq 0$ gives $y^T\mathbf{A}y \ge 0$. Hence

$$y^T\mathbf{A}y = 0 \qquad \text{for every } y \in \mathcal{C}(\mathbf{B}).$$

Step 3: from the quadratic form to the matrix. Since $\mathbf{A}$ is symmetric and non-negative definite it has a square root: $\mathbf{A} = L^TL$ (take $L = \mathbf{A}^{1/2} = P\Lambda^{1/2}P^T$ from the spectral decomposition). Then for $y \in \mathcal{C}(\mathbf{B})$,

$$0 = y^T\mathbf{A}y = y^TL^TLy = \|Ly\|^2 \quad\Longrightarrow\quad Ly = \mathbf{0} \quad\Longrightarrow\quad \mathbf{A}y = L^TLy = \mathbf{0}.$$

Step 4: conclude. Every column of $\mathbf{B}$ lies in $\mathcal{C}(\mathbf{B})$, so $\mathbf{A}$ annihilates each of them:

$$\mathbf{AB} = \mathbf{O}.$$

Transposing and using symmetry of both matrices, $\mathbf{O} = (\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T = \mathbf{BA}$. ■

Common confusion: "$y^T\mathbf{A}y = 0$" does not by itself give $\mathbf{A}y = 0$

For a general symmetric $\mathbf{A}$ it does not — e.g. $\mathbf{A} = \left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$ and $y = e_1$ give $y^T\mathbf{A}y = 0$ but $\mathbf{A}y = e_2 \ne \mathbf{0}$. Step 3 needs $\mathbf{A} \succeq 0$, which is exactly why that hypothesis is in the question. State this.

3 Practice: linear-algebra proofs at paper level

Five questions in the style of 2024 Q1/Q3 and 2025 Q4 — pure matrix algebra, no data. Q 3.1 is Homework 2 Q4 of this year, so it is the likeliest of the five.

Question 3.1. [**10** marks]

A is an $n \times n$ positive semidefinite matrix with $\mathrm{rank}(A) = r$ and $\mathrm{rank}(I_n - A) = n - r$, for some $1 \le r < n$. Prove that A is an orthogonal projection matrix. In addition, if $\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}, I_n)$ and $U_1 := \mathbf{Y}^TA\mathbf{Y}$, $U_2 := \mathbf{Y}^T\mathbf{Y} - \mathbf{Y}^TA\mathbf{Y}$, show — *without using the Fisher–Cochran theorem* — that U_1 and U_2 are independent, with $U_1 \sim \chi^2_r(\lambda_1/2)$ and $U_2 \sim \chi^2_{n-r}(\lambda_2/2)$, where $\lambda_1 = \boldsymbol{\mu}^TA\boldsymbol{\mu}$ and $\lambda_2 = \boldsymbol{\mu}^T\boldsymbol{\mu} - \boldsymbol{\mu}^TA\boldsymbol{\mu}$.

(Part 1: A is an orthogonal projector) A is positive semidefinite, hence symmetric, so spectrally decompose

$$A = P\Lambda P^T, \qquad P^TP = PP^T = I_n, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n),\ \lambda_i \ge 0.$$

- $\mathrm{rank}(A) = r \Longrightarrow$ **exactly** r of the λ_i are non-zero.

- $I_n - A = P(I - \Lambda)P^T$ has eigenvalues $1 - \lambda_i$, so $\operatorname{rank}(I_n - A) = n - r \implies$ exactly $n - r$ of the $1 - \lambda_i$ are non-zero $\implies$ **exactly** r of the λ_i equal 1.

Those r eigenvalues equal to 1 are certainly non-zero, and there are only r non-zero ones altogether. Hence the non-zero eigenvalues are *precisely* those r ones equal to 1:

$$\lambda_i \in \{0, 1\} \quad \forall i, \qquad \text{with exactly } r \text{ ones.}$$

Therefore $\Lambda^2 = \Lambda$, so $A^2 = P\Lambda^2 P^T = P\Lambda P^T = A$. Being symmetric and idempotent, A is an orthogonal projection matrix. ■

(Part 2: distributions and independence) Split $P = [\,C : D\,]$, where C is the $n \times r$ block of eigenvectors with eigenvalue 1 and D the $n \times (n - r)$ block with eigenvalue 0. Orthogonality of P gives

$$C^T C = I_r, \qquad D^T D = I_{n-r}, \qquad C^T D = 0, \qquad A = CC^T, \qquad I - A = DD^T.$$

Hence

$$U_1 = \mathbf{Y}^T CC^T \mathbf{Y} = \|C^T \mathbf{Y}\|^2, \qquad U_2 = \mathbf{Y}^T (I - A)\mathbf{Y} = \|D^T \mathbf{Y}\|^2.$$

Put $Z_1 = C^T \mathbf{Y}$ and $Z_2 = D^T \mathbf{Y}$. As linear images of a normal vector these are jointly normal, with

$$Z_1 \sim \mathcal{N}_r(C^T \boldsymbol{\mu},\ C^T C) = \mathcal{N}_r(C^T \boldsymbol{\mu}, I_r), \qquad Z_2 \sim \mathcal{N}_{n-r}(D^T \boldsymbol{\mu}, I_{n-r}),$$

$$\operatorname{Cov}(Z_1, Z_2) = C^T I_n D = C^T D = 0.$$

Jointly normal and uncorrelated $\implies$ independent; therefore $U_1 = \|Z_1\|^2$ and $U_2 = \|Z_2\|^2$ are independent. Finally, a squared norm of a $\mathcal{N}_k(\theta, I_k)$ vector is $\chi^2_k(\|\theta\|^2/2)$, and

$$\|C^T \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^T CC^T \boldsymbol{\mu} = \boldsymbol{\mu}^T A \boldsymbol{\mu} = \lambda_1, \qquad \|D^T \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^T (I - A)\boldsymbol{\mu} = \boldsymbol{\mu}^T \boldsymbol{\mu} - \boldsymbol{\mu}^T A \boldsymbol{\mu} = \lambda_2,$$

so $U_1 \sim \chi^2_r(\lambda_1/2)$ and $U_2 \sim \chi^2_{n-r}(\lambda_2/2)$. ■

Why "without Fisher–Cochran" is not a handicap

Once A is known to be a projector you can *split the space*: C and D are two blocks of a single orthogonal matrix, so independence is just $C^T D = 0$. Fisher–Cochran is the general r-piece theorem; this is the two-piece case done by hand, and it is shorter.

Question 3.2. [**8** marks]

Show that G is a generalised inverse of A if and only if AG is idempotent and $\operatorname{rank}(AG) = \operatorname{rank}(A)$.

($\Rightarrow$) Suppose $AGA = A$. Then

$$(AG)^2 = AGAG = (AGA)G = AG,$$

so AG is idempotent. For the rank, $\operatorname{rank}(AG) \le \operatorname{rank}(A)$ always, while $\operatorname{rank}(A) = \operatorname{rank}(AGA) \le \operatorname{rank}(AG)$. Hence $\operatorname{rank}(AG) = \operatorname{rank}(A)$.

($\Leftarrow$) Suppose AG is idempotent with $\operatorname{rank}(AG) = \operatorname{rank}(A)$. Since $\mathcal{C}(AG) \subseteq \mathcal{C}(A)$ and the two dimensions agree,

$$\mathcal{C}(AG) = \mathcal{C}(A).$$

An idempotent matrix acts as the identity on its own column space: if $u \in \mathcal{C}(AG)$, say $u = AGw$, then $(AG)u = (AG)^2 w = AGw = u$. Now take any x. Then $Ax \in \mathcal{C}(A) = \mathcal{C}(AG)$, so $Ax = AGu$ for some u, and

$$AGAx = AG(AGu) = (AG)^2 u = AGu = Ax.$$

This holds for every x, so $AGA = A$. ■

Common confusion: the rank condition cannot be dropped

AG idempotent alone is not enough: $G = O$ makes $AG = O$ idempotent, but $AGA = O \neq A$. The rank condition is what forces $\mathcal{C}(AG)$ to be all of $\mathcal{C}(A)$.

Question 3.3. [**6** marks]

For an $n \times p$ matrix $\mathbf{X}$, show that $\mathcal{N}(\mathbf{X}) = \mathcal{N}(\mathbf{X}^T\mathbf{X})$, and deduce that $\mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$ and $\operatorname{rank}(\mathbf{X}) = \operatorname{rank}(\mathbf{X}^T\mathbf{X})$.

Solution. $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}^T\mathbf{X})$**:** if $\mathbf{X}v = \mathbf{0}$ then $\mathbf{X}^T\mathbf{X}v = \mathbf{0}$.
$\mathcal{N}(\mathbf{X}^T\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X})$**:** if $\mathbf{X}^T\mathbf{X}v = \mathbf{0}$ then

$$0 = v^T\mathbf{X}^T\mathbf{X}v = \|\mathbf{X}v\|^2 \quad \Longrightarrow \quad \mathbf{X}v = \mathbf{0}.$$

So the two null spaces coincide. Both $\mathbf{X}$ and $\mathbf{X}^T\mathbf{X}$ have p columns, so rank–nullity gives

$$\operatorname{rank}(\mathbf{X}) = p - \dim\mathcal{N}(\mathbf{X}) = p - \dim\mathcal{N}(\mathbf{X}^T\mathbf{X}) = \operatorname{rank}(\mathbf{X}^T\mathbf{X}).$$

Finally $\mathcal{C}(\mathbf{X}^T\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^T)$ is immediate, and

$$\dim\mathcal{C}(\mathbf{X}^T\mathbf{X}) = \operatorname{rank}(\mathbf{X}^T\mathbf{X}) = \operatorname{rank}(\mathbf{X}) = \operatorname{rank}(\mathbf{X}^T) = \dim\mathcal{C}(\mathbf{X}^T),$$

so a subspace of equal dimension must be the whole space: $\mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$. ■

The two consequences to quote

- The normal equations $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{Y}$ are **always consistent**, because $\mathbf{X}^T\mathbf{Y} \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$.
- $a^T\boldsymbol{\beta}$ is estimable $\iff$ $a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$ — the two forms of the criterion are the same statement.

Question 3.4. [**10** marks]

Let G be any generalised inverse of $\mathbf{X}^T\mathbf{X}$ and set $P = \mathbf{X}G\mathbf{X}^T$. Show that P is the orthogonal projection matrix onto $\mathcal{C}(\mathbf{X})$, and that it does not depend on which g-inverse G is chosen.

(Step 1: $\mathbf{X}G\mathbf{X}^T\mathbf{X} = \mathbf{X}$) Put $M = \mathbf{X}G\mathbf{X}^T\mathbf{X} - \mathbf{X}$ and expand M^TM, using $\mathbf{X}^T\mathbf{X}G\mathbf{X}^T\mathbf{X} = \mathbf{X}^T\mathbf{X}$ repeatedly:

$$\begin{aligned} M^TM &= \mathbf{X}^T\mathbf{X}G^T\big(\mathbf{X}^T\mathbf{X}G\mathbf{X}^T\mathbf{X}\big) - \mathbf{X}^T\mathbf{X}G^T\mathbf{X}^T\mathbf{X} - \big(\mathbf{X}^T\mathbf{X}G\mathbf{X}^T\mathbf{X}\big) + \mathbf{X}^T\mathbf{X} \\ &= \mathbf{X}^T\mathbf{X}G^T\mathbf{X}^T\mathbf{X} - \mathbf{X}^T\mathbf{X}G^T\mathbf{X}^T\mathbf{X} - \mathbf{X}^T\mathbf{X} + \mathbf{X}^T\mathbf{X} = O. \end{aligned}$$

The diagonal entries of M^TM are the squared norms of the columns of M, so $M^TM = O$ forces $M = O$. Hence $\mathbf{X}G\mathbf{X}^T\mathbf{X} = \mathbf{X}$, i.e. $P\mathbf{X} = \mathbf{X}$.

(Step 2: idempotent) $P^2 = \mathbf{X}G\mathbf{X}^T \cdot \mathbf{X}G\mathbf{X}^T = \big(\mathbf{X}G\mathbf{X}^T\mathbf{X}\big)G\mathbf{X}^T = \mathbf{X}G\mathbf{X}^T = P$.

(Step 3: the image is $\mathcal{C}(\mathbf{X})$) $\mathcal{C}(P) \subseteq \mathcal{C}(\mathbf{X})$ from the leading factor $\mathbf{X}$; and $P\mathbf{X} = \mathbf{X}$ gives $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(P)$. So $\mathcal{C}(P) = \mathcal{C}(\mathbf{X})$.

(Step 4: invariance) Use the rule *AB^-C is invariant to the choice of B^- when $\mathcal{R}(A) \subseteq \mathcal{R}(B)$ and $\mathcal{C}(C) \subseteq \mathcal{C}(B)$*, with $A = \mathbf{X}$, $B = \mathbf{X}^T\mathbf{X}$, $C = \mathbf{X}^T$. By Q 3.3,

$$\mathcal{R}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) = \mathcal{R}(B), \qquad \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) = \mathcal{C}(B),$$

both with equality. Hence $P = \mathbf{X}G\mathbf{X}^T$ is the same matrix for every g-inverse G.

(Step 5: symmetry) A *symmetric* g-inverse of $\mathbf{X}^T\mathbf{X}$ always exists: spectrally decompose $\mathbf{X}^T\mathbf{X} = P_0\Lambda P_0^T$ and take $G_0 = P_0\Delta P_0^T$ with $\delta_i = 1/\lambda_i$ where $\lambda_i \neq 0$ and $\delta_i = 0$ elsewhere. For that choice $\mathbf{X}G_0\mathbf{X}^T$ is visibly symmetric — and by Step 4 every choice gives the *same* matrix. So P is symmetric.

Symmetric, idempotent, with image $\mathcal{C}(\mathbf{X})$: P is the orthogonal projector onto $\mathcal{C}(\mathbf{X})$. ■

The move worth stealing

Prove **invariance first**, then verify the awkward property (here, symmetry) for one conveniently chosen G. Trying to show $\mathbf{X}G\mathbf{X}^T$ is symmetric for an *arbitrary* G is painful and unnecessary.

Question 3.5. [**6** marks]

A and B are symmetric idempotent $n\times n$ matrices with $\mathcal{C}(B) \subseteq \mathcal{C}(A)$. Show that $AB = BA = B$, that $A - B$ is an orthogonal projection matrix, and that $\operatorname{rank}(A-B) = \operatorname{rank}(A) - \operatorname{rank}(B)$.

Solution. $AB = B$**.** A is symmetric and idempotent, hence the orthogonal projector onto $\mathcal{C}(A)$, so $Au = u$ for every $u \in \mathcal{C}(A)$. For any x we have $Bx \in \mathcal{C}(B) \subseteq \mathcal{C}(A)$, so $A(Bx) = Bx$. As this holds for all x, $AB = B$.

$BA = B$**.** Transpose, using symmetry of both: $BA = B^TA^T = (AB)^T = B^T = B$.

$A - B$ **is an orthogonal projector.** It is symmetric as a difference of symmetric matrices, and

$$(A-B)^2 = A^2 - AB - BA + B^2 = A - B - B + B = A - B.$$

Rank. $A - B$ is idempotent, so trace counts rank:

$$\operatorname{rank}(A-B) = \operatorname{tr}(A-B) = \operatorname{tr}(A) - \operatorname{tr}(B) = \operatorname{rank}(A) - \operatorname{rank}(B). \quad ■$$

Where you have already used this

This is exactly why $P_{\mathbf{X}} - P_{\mathbf{Z}}$ is a projector of rank $\operatorname{rank}(\mathbf{X}) - \operatorname{rank}(\mathbf{Z})$ in the nested-model decomposition $P_{\mathbf{X}} = P_{\mathbf{Z}} + P_{\mathbf{W}|\mathbf{Z}}$ — and hence why $\mathrm{SSE}_{red} - \mathrm{SSE}_{full}$ is a chi-square with the right degrees of freedom. Every ANOVA table rests on it.

The five moves these questions keep testing

1. Eigenvalues of an idempotent matrix lie in $\{0, 1\}$, hence $\operatorname{tr} = \operatorname{rank}$. (Q 3.1, Q 3.5)
2. Spectral decomposition to split $\mathbb{R}^n$ into orthogonal blocks. (Q 3.1)
3. $M^TM = O \Rightarrow M = O$, and $v^TAv = \|\mathbf{X}v\|^2 = 0 \Rightarrow \mathbf{X}v = \mathbf{0}$. (Q 3.3, Q 3.4)
4. An idempotent matrix acts as the identity on its own column space. (Q 3.2, Q 3.5)
5. Prove invariance first, then check the property for one convenient choice. (Q 3.4)

If a question asks you to prove anything about projections, the first line is "symmetric and idempotent" and the second is $\operatorname{tr} = \operatorname{rank}$.

4 Coverage audit: do the crash courses cover all of this?

They do now. When these solutions were first written, the machinery for every question was in the crash courses but four specific results that were actually asked were not. Those four have since been **patched into Crash Courses 2 and 3**, and the proof of the third fundamental theorem has been filled in. The table below records where each one now lives.

Paper	Q	Where the method is	Status
2024	1	CC2 §2.4 *Sums and differences of projections*	**Added**
2024	2	CC2 §4–5 (estimability, G–M); CC3 §6 (test)	Already covered
2024	3	CC2 §2.8 *Building a g-inverse from a full-rank corner*	**Added**
2024	4	CC3 §2.4(a),(b) *Two consequences*	**Added**
2025	1	CC3 §7 (3rd fundamental theorem, proof now given)	**Strengthened**
2025	2	CC2 §4.2 (the 2×2 $\alpha_i + \gamma_j$ example)	Already covered
2025	3	CC3 §2.4(c) *Analysis of covariance*	**Added**
2025	4	CC2 §2.5 *A non-negative-definite squeeze*	**Added**

The four results that were missing, and where they now are

1. **$\mathbf{P} + \mathbf{Q}$ is a projector $\iff$ $\mathbf{PQ} = \mathbf{O}$**, with rank additivity (2024 Q1). Now **CC2 §2.4**, together with the mirror result that $\mathbf{P} - \mathbf{Q}$ is a projector when $\mathcal{C}(\mathbf{Q}) \subseteq \mathcal{C}(\mathbf{P})$ — which is what makes $P_{\mathbf{X}} - P_{\mathbf{Z}}$ a projector in the nested-model decomposition.
2. **The block g-inverse $G = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$** (2024 Q3). Now **CC2 §2.8**, with the point that the rank hypothesis is used in exactly one place: forcing the Schur complement to vanish.
3. **$\mathbf{A} \succeq 0$, $I - \mathbf{A} - \mathbf{B} \succeq 0$, $\mathbf{B}^2 = \mathbf{B} \Rightarrow \mathbf{AB} = \mathbf{BA} = \mathbf{O}$** (2025 Q4). Now **CC2 §2.5**, with the counterexample showing why $\mathbf{A} \succeq 0$ cannot be dropped.
4. $\mathrm{Var}(\widehat{\beta}_j) = \sigma^2/d_j$**, the T_j decomposition and the collinearity argument** (2024 Q4), and the **ANCOVA** instance of Frisch–Waugh with the $S_{zz}, S_{zw}, \ldots$ notation and the U, V statistics (2025 Q3). Now **CC3 §2.4**, parts (a)–(c).

What the two papers never asked

No numerical ANOVA table, no calculator, no confidence intervals, no prediction, no R^2. Both papers were pure theory. **This year's guide breaks that pattern** — it promises a numerical problem and three hours instead of two — so Crash Course 4 (one-way ANOVA) and §10 of the formula sheet cover ground the past papers do not test. Prepare both.

Companions: the four crash courses, and the two-page formula sheet.